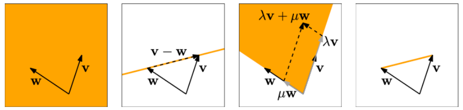
1 Part 1

1.1 Introduction

Affine, Conic, and Convex Linear Combinations $\lambda_1 v_1 + ... + \lambda_n v_n$ is:
- Affine if $\sum_{i=1}^{n} \lambda_i = 1$
- Conic if $\forall j \in [n], \lambda_j > 0$
- Convex if both affine and conic.

between. For $\lambda < 0$ we end up left of **w**, and for $\lambda > 1$, we will be right of **v**. Generally, any point on the line is obtained by the rule "go to **w**, then add a scalar multiple of **v**−**w**!". See Figure 1.9 (ii).



(i) linear (ii) affine (iii) conic (iv) convex

combinations of two vectors

Notation Example: $(i^2)_{i=1}^{m}$ constructs $(1, 4, ..., m^2)^\top$.
For matrices:

$$A = [a_{ij}]_{i=1,j=1}^{m,n}$$

The Kronecker Delta, δ_{ij} is defined as 1 if $i = j$ or 0 else.

Dot Products $v \cdot w = v^\top w = \langle v, w \rangle$
- $v^\top w = w^\top v$
- $(\lambda v)^\top w = \lambda(v^\top w)$
- $u^\top(v + w) = u^\top v + u^\top w$ (left and right distributivity)
- $v^\top v = 0 \iff v = 0$

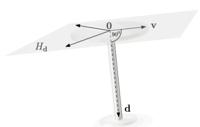
Euclidean Norm $\|v\|_2 = \sqrt{v^\top v}$

Unit Vector $\frac{v}{\|v\|}$

Cauchy-Schwarz Inequality $|v^\top w| \leq \|v\| \, \|w\|$, whereby equality holds iff a vector is a scalar multiple of the other.

Cosine $\cos(\alpha) = \frac{v^\top w}{\|v\| \, \|w\|}$ **TODO: Add Figure 1.17?**

Hyperplane through Origin Let $d \in \mathbb{R}^m, d \neq 0$, then $H_d = \{v \in \mathbb{R}^m : v^\top d = 0\}$.



Triangle Inequality $\|v + w\| \leq \|v\| + \|w\|$

Span Set of all linear combinations. If v_{k+1} is a linear combination of the previous, then $\mathrm{Span}(v_1, ..., v_k) = \mathrm{Span}(v_1, ..., v_k, v_{k+1})$.
The span of m linearly independent vectors is $\mathbb{R}^m$.
Matrices: $C(A) = \{Ax : x \in \mathbb{R}^n\}$ and the column space of a matrix is unchanged by adding or removing dependent columns.

1.2 Linear Independence

Vectors are linearly dependent if:
- at least one of them is a linear combination of the others/previous, i.e.:

$$\exists k \in [n], v_k = \sum_{j=1, j \neq k}^{n} \lambda_j v_j$$

- 0 is a nontrivial linear combination of the vectors

The columns of a matrix A are linearly independent iff $x = 0$ is the only x such that $Ax = 0$.

A set G is linearly dependent if there is a $v \in G$ such that it is a linear combination of $G \setminus \{v\}$.

1.3 Matrices

If $\forall i, j, a_{ij} = a_{ji}$ then A is a symmetric matrix.

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 4 & 7 \\ 0 & 7 & 5 \end{bmatrix}$$

symmetric matrix

A vector $b \in \mathbb{R}^m$ is in $C(A) \iff \exists x \in \mathbb{R}^n, Ax = b$.

Rank-1 Matrices

$$\mathrm{rank}(A) = 1 \iff \mathrm{rank}(A^\top) = 1 \iff \exists v \in \mathbb{R}^m, w \in \mathbb{R}^n, A = [v_i w_j]_{i=1,j=1}^{m,n}$$

Linearity $T(\lambda_1 x_1 + \lambda_2 x_2) = \lambda_1 T(x_1) + \lambda_2 T(x_2)$ or, split:
- $T(x + x') = T(x) + T(x')$, e.g., $A(x + x') = Ax + Ax'$
- $T(\lambda x) = \lambda T(x)$, e.g., $A(\lambda x) = \lambda Ax$

Every matrix transformation is linear.

Kernel and Nullspace $\ker(T) = \{x \in \mathbb{R}^n : T(x) = 0\} = N(A_T)$

Image $\mathrm{im}(T) = \{T(x) : x \in \mathbb{R}^n\} = C(A_T)$

Transpose $(ABC)^\top = C^\top B^\top A^\top$

Distributivity, Associativity $A(B + C) = AB + AC$, $(A + B)C = AC + BC$ and $(AB)C = A(BC)$

CR Decomposition For a matrix A, we can decompose it into $A = CR'$, where C contains the independent columns and R' the multiples. E.g.,

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

whereby C has full column and R' row rank.

Function Types
- Injective if for every $y \in Y$, where is at most one $x \in X$ with $f(x) = y$
- Surjective if for every $y \in Y$, there is at least one $x \in X$ with $f(x) = y$
- Bijective if both.

Wide matrices are not injective and tall matrices not surjective.

Thus, for a matrix $A \in \mathbb{R}^{m \times m}$:

$$T_A \text{ bijective} \iff \exists B, BA = I \iff \text{Columns of } A \text{ linearly independent}$$

Invertibility A invertible $\iff \exists B, BA = BA = I \iff A$ full column rank

$(A^\top)^{-1} = (A^{-1})^\top$

$(AB)^{-1} = B^{-1} A^{-1}$

Gauss Elimination Produces an UT matrix. If a zero pivot is hit and row exchange is not possible, the algorithm fails and the columns are linearly dependent. One can run this on a right hand side matrix B to solve for multiple bs.

Invariance Let $A \in \mathbb{R}^{m \times n}, M \in \mathbb{R}^{m \times m}$ with M invertible. Then:
- $Ax = b$ and $MAx = Mb$ have the same solutions x.
- $N(A) = N(MA)$
- A linearly independent $\iff MA$ linearly independent
- $R(A) = R(MA)$

RREF A matrix is in RREF is there is some natural number $r \leq m$ and column indices $1 \leq j_1 < j_2 < ... j_r \leq n$ such that the following conditions hold:
- For every $i \in [r]$, column j_i of R is the standard unit vector e_i.
- All entries r_{ij} below the staircase are zero.

We also have:
- A matrix R in $\mathrm{RREF}(j_1, j_2, ..., j_r)$ has independent columns $j_1, j_2, ..., j_r$ and therefore rank r.
- If some $c_i \neq 0$ for $i > r$, then $Rx = c$ has no solutions.
- Uniqueness of RREF solution

1.3.1 Computing Inverses

- Converting A to RREF with RHS I. A^{-1} is the elimination matrix.

1.4 Vector Spaces

Definition 4.1 (Vector space). *A vector space is a triple $(V, +, \cdot)$ where V is a set (the vectors), and*

$$+ \;:\; V \times V \to V \quad \textit{is a function (vector addition),}$$
$$\cdot \;:\; \mathbb{R} \times V \to V \quad \textit{is a function (scalar multiplication),}$$

satisfying the following axioms of a vector space *for all* $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ *and all* $\lambda, \mu \in \mathbb{R}$.

1. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ — *commutativity*
2. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ — *associativity*
3. *There is a vector* $\mathbf{0}$ *such that* $\mathbf{v} + \mathbf{0} = \mathbf{v}$ *for all* $\mathbf{v}$ — *zero vector*
4. *There is a vector* $-\mathbf{v}$ *such that* $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ — *negative vector*
5. $1 \cdot \mathbf{v} = \mathbf{v}$ — *identity element*
6. $(\lambda \cdot \mu)\mathbf{v} = \lambda \cdot (\mu \cdot \mathbf{v})$ — *compatibility of · and · in* $\mathbb{R}$
7. $\lambda(\mathbf{v} + \mathbf{w}) = \lambda\mathbf{v} + \lambda\mathbf{w}$ — *distributivity over +*
8. $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$ — *distributivity over + in* $\mathbb{R}$

Every vector space V contains exactly one zero vector. For every $v \in V$, there is exactly one $-v$.

Subspace Axioms Given $v, w \in U$:
- $v + w \in U$
- $\lambda v \in U$ and thus $0 \in U$

And thus we also have closure under linear combination.

Fundamental Subspaces
- $C(A), R(A), N(A)$

Finitely Generated Vector Space if there is a finite subset $G \subseteq V$ with $\mathrm{Span}(G) = V$. If a $G \subseteq V$ has $\mathrm{Span}(G) = V$, it forms a basis.

Steinitz Exchange Lemma Let V be a finitely generated vector space, $F \subseteq V$ a finite set of linearly independent vectors, and $G \subseteq V$ a finite set of vectors with $\mathrm{Span}(G) = V$. Then:
- $|F| \leq |G|$
- There is a $E \subseteq G$, $|E| = |G| - |F|$ such that $\mathrm{Span}(F \cup E) = V$

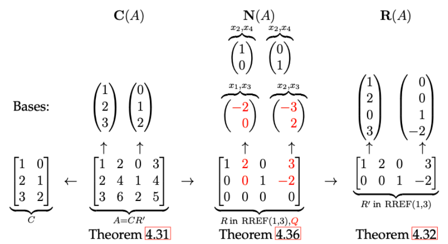
Dimension $\dim(V) = |B|$, where B is an arbitrary basis of the vector space.

Isomorphism All m-dimensional vector spaces are isomorphic.

Dimensions $\dim(C(A)) = \mathrm{rank}(A) = r = \dim(R(A))$

Lemma *(4.35: Nullspace Isomorphism)* $\dim(N(R)) = n - r$. The linearly dependent columns obtained from RREF form a basis of the nullspace. There is an isomorphism from $N(N) \to \mathbb{R}^{n-r}$.

Summary. Here a visual summary (on our running example). It shows the computation of (bases of) the three fundamental subspaces of A (column space, null space, row space), based on Gauss-Jordan elimination applied to A (Theorem 3.17). The CR decomposition $A = CR'$ (Theorem 2.46) is obtained as a by-product (Theorem 3.18).



Solution Space The set $\mathrm{Sol}(A, b) = \{x \in \mathbb{R}^n : Ax = b\} \subseteq \mathbb{R}^n$ is the solution space of $Ax = b$. If $b \neq 0$, this is not a subspace as the zero vector is not included.

This can be remedied by defining $\mathrm{Sol}(A, b) = \{s + x : x \in N(A)\}$ to get an affine subspace.

$\dim(\mathrm{Sol}(A, b)) = \dim(N(A))$

Affine Subspace Contains points whose coordinates are absolute as opposed to defined from the origin. Points do not add or scale.

2 Part 2

2.1 Orthogonality

Subspaces V, W are orthogonal $\iff$ all their bases v_i, w_i are orthogonal. Similarly, if V, W are two orthogonal subspaces of $\mathbb{R}^n$, the set of their bases is linearly independent and $V \cap W = \{0\}$, $\dim(V + W) = \dim(V) + \dim(W) \leq n$.

Per Theorem 5.1.7, we have the following equivalences:
- $W = V^\perp$
- $\dim(V) + \dim(W) = n$
- Every $u \in \mathbb{R}^n$ can be written as $u = v + w, v \in V, w \in W$

Orthogonal Complement $V^\perp = \{w \in \mathbb{R}^n : w^\top v = 0 \forall v \in V\}$. We have $(V^\perp)^\perp = V$.
- $N(A) = C(A^\top)^\perp = R(A)^\perp$
- $N(A) = C(A^\top)^\perp$ and $C(A^\top) = N(A)^\perp$
- $N(A) = N(A^\top A)$ and $C(A^\top) = C(A^\top A)$

2.2 Projections

A projection onto a subspace S is the $\arg\min_{p \in S} \|b - p\|$, i.e., point closest to the original vector.

Projection onto a line λa Well-defined; $\mathrm{proj}_{S(b)} = \frac{a a^\top}{a^\top a} b$

Projection onto $C(A)$ Well-defined; $\mathrm{proj}_{S(b)} = A\hat{x}$ where $A^\top A \hat{x} = A^\top b$

Alternatively, one could calculate the projection matrix $P = A(A^\top A)^{-1} A^\top$ and use $\mathrm{proj}_{S(b)} = Pb$

Invertibility of $A^\top A$ invertible $\iff A$ has linearly independent columns

2.3 Set of all Solutions to a System

Lemma *(6.2.1)* Let $x, y \in C(A^\top)$ $Ax = Ay \iff x = y$

If $\{x \in \mathbb{R}^n : Ax = b\} \neq \emptyset$, then $\{x \in \mathbb{R}^n : Ax = b\} = x_1 + N(A), x_1 \in R(A)$, where x_1 is unique such that $Ax_1 = b$. As a corollary, if this set is not empty, there is a unique $x_1 \in C(A^\top A)$ such that $Ax_1 = b$.

Unsatisfiability Proof $\{x \in \mathbb{R}^n : Ax = b\} = \emptyset \iff \{z \in \mathbb{R}^m : A^\top z = 0, b^\top z = 1\} \neq \emptyset$ – i.e., find a suitable z to satisfy this.

2.4 Orthonormalization

Vectors are orthonormal if $\forall i, j \in [n], q_i^\top q_j = 0$ unless $i = j$ and have norm 1.

Orthogonal matrices preserve norm and inner product. i.e., $\|Qx\| = \|x\|, (Qx)^\top (Qy) = x^\top y$

If the columns of Q are an orthonormal basis for S, then the projection matrix onto S is given by $QQ^\top$ and the least squares solution to $Qx = b$ by $\hat{x} = Q^\top b$.

GSO Returns an orthonormal basis of the input vectors.
- $q_1 = \frac{a_1}{\|a_1\|}$
- $$q_{k'} = a_k - \sum_{i=1}^{k-1} (a_k^\top q_i) q_i$$

  and $q_k = \frac{q_{k'}}{\|q_{k'}\|}$

QR Decomposition We can decompose $A = QR$ such that Q is the output of GSO and R is an upper triangular matrix given by $R = Q^\top A$.
Note that R is invertible and that we have $QQ^\top A = A$.

We can use QRD for projections and least squares:
- $\mathrm{proj}_{C(A)}(b) = QQ^\top b$
- The least squares solution for $Ax = b$, $\hat{x}$, can be calculated using: $R\hat{x} = Q^\top b$.

2.5 Pseudoinverses

Full Column Rank $A^\dagger = (A^\top A)^{-1} A^\top$; $A^\dagger$ is a left inverse such that $A^\dagger A = I$.

Full Row Rank $A^\dagger = A^{\top (AA^\top)^{-1}}$; $A^\dagger$ is a right inverse such that $AA^\dagger = I$

General $A^\dagger = R^\dagger C^\dagger$ or $A^\dagger = R^{\top (C^\top A R^\top)^{-1}} C^\top$ (same applies to any full-row, full-col decomposition per P6.4.9)

Equalities
- $AA^\dagger A = A$ and $A^\dagger A A^\dagger = A^\dagger$ and $(A^\top)^\dagger = (A^\dagger)^\top$
- If $AA^\dagger$ is symmetric, it is the projection matrix onto $C(A)$
- If $A^\dagger A$ is symmetric, it is the projection matrix onto $C(A^\top) = R(A)$

2.6 Determinant

Do not forget multilinearity! See examples. Defined for square matrices. For finding the inverse of a 2×2 matrix: $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$.

Axiomatically:

1. Linear in each column (multilinear)
2. $\det(I) = 1$
3. $\det(A) = 0$ if A has two identical columns

$\det(AW) = \det(A)\det(W)$

A invertible $\iff \det(A) \neq 0$

Sign

$$\operatorname{sgn}(\sigma) = \begin{cases} 1 \text{ if } & |(i,j)| : i < j \land \sigma(i) > \sigma(j) \text{ even} \\ -1 \text{ if odd} \end{cases}$$

sgn is multiplicative: $\operatorname{sgn}(\sigma \circ \gamma) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\gamma)$

$\forall n \geq 2$, exactly half of permutations have sign 1, other half -1.

And:

- $\det(A^\top) = \det(A)$
- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = \frac{1}{\det(A)}$
- When swapping two elements of A to get A', then $\det(A') = -\det(A)$

Determinant of:

- 2×2: $ad - bc$
- 3×3: $aei + bfg + cdh - ceg - bdi - afh$
- Triangular Matrices: $\det(T) = \prod_{k=1}^n T_{kk}$
- Permutation Matrices: $\det(P) = \operatorname{sgn}(\sigma) = \operatorname{sgn}(P)$
- Orthogonal Matrices: $\det(Q) \in \{1, -1\}$

Inverse $A^{-1} = \frac{1}{\det(A)} C^\top$ or $AC^\top = \det(A) I$ (inefficient)

Cramer's Rule Given $A \in \mathbb{R}^{n \times n}$, $\det(A) \neq 0$, the jth row of the solution is given by: $x_j = \frac{\det(\mathcal{B})}{\det(A)}$. $\mathcal{B}$ is the matrix obtained by replacing the jth column of A with the vector b. (inefficient)

2.6.1 Laplace Expansion

$$\det(A) = (-1)^{i+j} A_{i,j} M_{i,j}$$

In other words, for every entry in the first row:

- Eliminate its row, column and calculate its determinant
- Multiply this determinant by the eliminated entry and $(-1)^{1+j}$

Then sum.

2.7 Complex numbered-units

- $|a + bi| = \sqrt{a^2 + b^2}$ (Modulus)
- $\overline{a + bi} = a - bi$ (Conjugate)
- $\frac{1}{z} = \frac{\overline{z}}{|z|^2}$
- $e^{i\theta} = \cos(\theta) + i\sin(\theta)$, i.e., $z = re^{i\theta}$, where r is the modulus of z.

Fundamental Theorem of Algebra $\mathbb{C}$ is algebraically closed, i.e., a polynomial of degree n has n roots in $\mathbb{C}$.

The number of times a root is repeated is its **algebraic multiplicity**. Note that **geometric multiplicity** is $\dim(N(A - \lambda I))$.

2.8 Eigenvalues and Eigenvectors

Given an $A \in \mathbb{R}^{n \times n}$:

- $Av = \lambda v$ (or $(A - \lambda I)v = 0$), where v is the eigenvector and λ the eigenvalue.
- λ is a real eigenvalue of an $A \iff \det(A - \lambda I) = 0$. A nonzero vector v is an eigenvector associated with $\lambda \iff v \in N(A - \lambda I)$
- $\det(A - \lambda I)$ is a polynomial of degree n and the coefficient of the λ^n term is $(-1)^n$
- Every square matrix has an eigenvalue
- If λ is an EV of A, A^k is an EV of A^k – i.e., $\frac{1}{\lambda}$ and v are an EV pair for A^{-1}.
- If A has n distinct eigenvalues, the EVs are linearly independent and one can build a basis of $\mathbb{R}^n$
- The eigenvalues of $A = A^\top$, but not eigenvectors.

If (λ, v) is an eigenvalue pair, $(\overline{\lambda}, \overline{v})$ is an eigenvalue pair.

Gaussian elimination does not preserve EVs.

EVs of:

- **Triangular Matrices** Eigenvalues: Entries in the diagonal
- **Diagonal Matrices** Eigenvalues: Entries in the diagonal. Eigenvectors: Basis vectors $e_1, \ldots, e_n$.

Trace $\operatorname{Tr}(A) = \sum_{i=1}^n A_{ii}$. The trace is equal to the sum of the eigenvalues – $\operatorname{Tr}(A) = \sum_{i=1}^n \lambda_i$.

Further, $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$ and $\operatorname{Tr}(ABC) = \operatorname{Tr}(BCA) = \operatorname{Tr}(CAB)$

Eigenvalues of Orthogonal Matrices If λ is an eigenvalue of Q, then $|\lambda| = 1$

Characteristic Polynomial $\det(zI - A)$

2.9 Symmetric Matrices and Spectral Theorem

$A = V\Lambda V^{-1}$, where Λ is a diagonal matrix with

- Λ_{ii} equal to the ith eigenvalue λ_i and 0 otherwise
- V having the eigenvectors as columns

Diagonalizability Whether or not there is an invertible V such that $V^{-1}AV = \Lambda$.

A matrix is orthogonally diagonalizable $\iff D^\top = D$ (symmetric). Proof: $D = V\Lambda V^{-1} \implies D = V\Lambda V^\top \implies D^\top = (V\Lambda V^\top)^\top = V\Lambda^\top V^\top = V\Lambda V$ [Λ diagonal] $= D$

Eigenvalues of Projections If P is a projection matrix on $U \subseteq \mathbb{R}^n$, then P has eigenvalues 0 and 1, and a complete set of real eigenvectors.

Similar Matrices have the same eigenvalues and trace. A has a complete set of eigenvectors $\iff B$ does.

Complete set of Eigenvectors If an $A \in \mathbb{R}^{n \times n}$ has enough eigenvectors to build a basis of $\mathbb{R}^n$. A matrix has a complete set of **real** eigenvectors if all its eigenvalues are real and the geometric multiplicities = algebraic multiplicities for all eigenvalues.

2.10 Spectral Theorem

Any symmetric matrix $A \in \mathbb{R}^{n \times n}$ has n real eigenvalues and an orthonormal basis of $\mathbb{R}^n$ consisting of its eigenvectors.

As a result, for any symmetric matrix, there exists an orthogonal matrix V (whose columns are the EVs of A) such that $A = V\Lambda V^\top$.

The rank of a real symmetric matrix is the number of nonzero eigenvalues, counting repetitions.

Given an orthonormal basis of eigenvectors of A: $v_1, \ldots, v_n$ and associated values $\lambda_1, \ldots, \lambda_n$, we have: $A = \sum_{k=1}^n \lambda_i v_i v_i^\top$.

Rayleigh Quotient For a symmetric matrix and $x \neq 0$, $R(x) = \frac{x^\top Ax}{x^\top x}$ has a maximum at $R(v_{\max}) = \lambda_{\max}$ and minimum $R(v_{\min}) = \lambda_{\min}$ whereby $\lambda_{\max}, \lambda_{\min}$ are the largest and smallest eigenvalues of A, v associated eigenvectors.

Positive Semidefinite A symmetric matrix is PSD if all its eigenvalues are ≥ 0. Positive definite is > 0. An A is PSD $\iff \forall x \in \mathbb{R}^n, x^\top Ax \geq 0$ and for PD, > 0.

PSD and PD matrices are closed under addition. A, B P(S)D $\implies A + B$ P(S)D

Gram Matrices $G_{ij} = v_i^\top v_j$ or $AA^\top$.

Given a real matrix A, the nonzero eigenvalues of $A^\top A = AA^\top$. Both are symmetric and semidefinite.

Cholesky Decomposition Every symmetric PSD M is a Gram matrix of an UT matrix C. $M = C^\top C$ is the Cholesky decomposition.

2.10.1 SVD

A **singular** matrix is a square matrix that is not invertible ($\det(A) = 0$).

Given $A \in \mathbb{R}^{m \times n}$. $\exists U \in \mathbb{R}^{m \times m}, \exists V \in \mathbb{R}^{n \times n}, A = U\Sigma V^\top$, where Σ is diagonal and the diagonal elements are non-negative and ordered in descending order. $U^\top U = V^\top V = I$ (orthonormal)

The columns of U are the left, columns of V the right singular vectors. The diagonal elements of Σ are the singular values of A.

If A has rank r, we can write $A = U_r \Sigma_r V_r^\top$, where U_r contains the first r left singular vectors, analog for V, Σ.

We have $AA^\top = U(\Sigma\Sigma^\top)U^\top$, i.e., columns of U are eigenvectors of $AA^\top$, singular values of A the square root of the eigenvalues of $AA^\top$. If $m > n$, A has n singular values, $AA^\top$ has m, so the missing ones are filled with zeroes.

Every matrix A has an SVD, i.e., every linear transformation is diagonal when viewed in the bases of the singular vectors.

Lastly, $A = \sum_{k=1}^r \sigma_k u_k v_k^\top$ – every matrix can be expressed as a sum of rank-1 matrices.

3 Miscellaneous

Rotation matrix in $\mathbb{R}^{2 \times 2} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Closed Form Expression in only constants, variables, and elementary operations like $+, -, \times, /$, and integer powers and function composition.

4 Tips

4.1 Examples

4.1.1 Multilinearity of Determinant

Given $A \in \mathbb{R}^{3 \times 3}$, $\det(A) = 2$, find $\det(2A)$.

Because the determinant is multilinear, use the formula $\det(cA) = c^n \det(A)$. So 16.

4.1.2 Guessing Eigenvectors for Recurrences

For recurrences $g_n = \begin{pmatrix} a_{n+2} \\ a_{n+1} \\ a_n \\ \ldots \end{pmatrix}$, the eigenvectors are usually:

$$\begin{pmatrix} \lambda^{k-1} \\ \lambda^{k-2} \\ \vdots \\ \lambda \\ 1 \end{pmatrix}$$

4.1.3 Fibonacci Sequence

Given $F_0 = 0, F_1 = 1$, then $F_n = F_{n-1} + F_{n-2}$, we can write: $\underbrace{\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix}}_{g_n} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}}_{M} \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}$, and thus: $g_n = M^n g_0$

Closed Form Expression

$$M = \begin{pmatrix} 5 & -6 \\ 1 & 0 \end{pmatrix}, g_0 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, g_n = \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix}$$

So:

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = \begin{bmatrix} 5 & -6 \\ 1 & 0 \end{bmatrix}^n \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

3.1 Eigenvalues and -Vectors

$$\begin{vmatrix} 5 - \lambda & -6 \\ 1 & -\lambda \end{vmatrix} = 0 \implies -\lambda(5 - \lambda) + 6 = 0 \implies \lambda^2 - 5\lambda + 6 = 0 \implies (\lambda - 2)(\lambda - 3) = 0 \implies \lambda \in \{2, 3\}$$

Vector for $\lambda = 2$

$$\begin{bmatrix} 5 - 2 & -6 \\ 1 & 0 - 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \implies \begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

Setting $x_2 = 1$: $x_1 - 2 = 0 \implies x_1 = 2$. So $\binom{2}{1}$ is a basis vector for $\lambda = 2$.

Vector for $\lambda = 3$

$$\begin{bmatrix} 5 - 3 & -6 \\ 1 & 0 - 3 \end{bmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 0 \implies \begin{bmatrix} 2 & -6 \\ 1 & -3 \end{bmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 0$$

Setting $y_2 = 1$: $2y_1 - 6 = 0 \implies y_1 = 3$. So $\binom{3}{1}$ is a basis vector for $\lambda = 3$.

3.2 Closed Form Expression

Writing g_0 in terms of the eigenbasis: $g_0 = \binom{2}{0} = 2\binom{3}{1} - 2\binom{2}{1}$

So $g_n = 2\left(M^n\binom{3}{1} - M^n\binom{2}{1}\right) \implies g_n = 2\left(3^n\binom{2}{1} - 2^n\binom{2}{1}\right)$.

And thus, by only considering the second row in the equation:

$$\boldsymbol{a_n = 2(3^n - 2^n)}$$

4.2 Algorithms

4.2.1 Gram-Schmidt

We start with the basis:

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

4.2.1.1 Step 1: Get first unit vector q_1

Normalize v_1 immediately to get the first orthonormal vector q_1:

$$r_{\{11\}} = \|v_1\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$q_1 = \frac{v_1}{r_{\{11\}}} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

4.2.1.2 Step 2: Get second unit vector q_2

Now we calculate the projection of v_2 onto q_1. Since q_1 is already a unit vector, we use the simpler formula (no division):

$$p = (q_1^T v_2) q_1$$

Calculate the dot product scalar $(q_1^T v_2)$:

$$q_1^T v_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}^T \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$= \left(\frac{1}{\sqrt{2}} \cdot 0\right) + \left(\frac{1}{\sqrt{2}} \cdot 2\right) = \frac{2}{\sqrt{2}} = \sqrt{2}$$

Multiply this scalar by the vector q_1:

$$p = \sqrt{2} \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Subtract this projection from v_2 to find the perpendicular part w_2:

$$w_2 = v_2 - p$$

$$= \begin{pmatrix} 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Finally, normalize w_2 to get q_2:

$$r_{\{22\}} = \|w_2\| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$q_2 = \frac{w_2}{r_{\{22\}}} = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

4.2.1.3 Result

$$q_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad q_2 = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

4.3 SVD

Compute $A^\top A$. Solve $\det(A^\top A - \lambda I) = 0$ to find Evals. Take `sqrt` of Evals for Σ.

Find eigenvectors of $A^\top A$ via $\det(A^\top A - \lambda_i I) = 0$. This gives the right singular vectors (V).

Then, solve $u_i = \frac{1}{\sigma_i} Av_i$ to find Evectors of U.

4.4 Gaussian Elimination

TODO

4.5 Least Squares

TODO

5 Exercise Results

5.1 Sheet 13

- If $S^\top = -S$, $-S^2$ is symmetric and PSD.
- A symmetric diagonally dominant matrix (for every row, abs value of the diagonal entry is at least the sum of other abs values in that row) $\in \mathbb{R}^{n\times n}$ with non-negative diagonal entries is PSD.
- $A^\dagger = V_r \Sigma_r^{-1} U_r^\top$
- Given A with $\operatorname{rank}(A) = r$, its SVD, and $c = U^\top b$, we have $\min_{x\in\mathbb{R}^n} \|Ax - b\|_2^2 = \min_{y\in\mathbb{R}^n} \|\Sigma y - c\|_2^2$. The opt y is:

$$y^* = \begin{pmatrix} \frac{c_1}{\sigma_1} \\ \vdots \\ \frac{c_r}{\sigma_r} \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \arg\min_{y\in\mathbb{R}^n} \|\Sigma y - c\|_2^2$$

5.2 Sheet 12

- If eV of $A \iff$ eV of B, then $AB = BA$

5.3 Sheet 12

Given a block matrix $M = \begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$, $\det(M) = \det(A)\det(C)$

5.4 Sheet 9

- There is a bijection $C(A) \to R(A)$